Cai Chen

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Education

Iowa State University Ames, Iowa
Bachelor of Science in Computer Engineering *Expected December 2026*

Experience

Founder and Software Engineer February 2025 – Present
Chardle.com *Remote*

- Built and scaled a Next.js web platform to **120,000+ monthly active users**, shipping core content, then iterating by fixing bugs and adding features such as new gamemodes, multiplayer, and social features.
- Cut infrastructure costs by **90%** by migrating to AWS architecture (Lambda, API Gateway, DynamoDB, S3).
- Engineered stateful WebSocket infrastructure via Cloudflare Durable Objects, achieving 1.04ms average CPU time and zero errors across 30,000+ daily real-time multiplayer requests.
- Grew monthly revenue by **120%** by negotiating an ad-tech partnership with Venatus, optimizing ad inventory.

Software Developer Intern June 2026 – August 2026
W.R. Berkley *Urbandale, Iowa*

- Contributed to a **1.3M+ line of code** enterprise codebase maintained by 50+ engineers, boosting **development velocity by 32%** within a dedicated 6-member team.
- Refactored email micro-service, eliminating redundant code across 10 modules, implementing strict validation and linting to improve long-term maintainability and readability for **Fortune 500 company**.
- Architected an agentic AI workflow integrating custom autonomous agents with human-in-the-loop validation, **saving 50 hours** of development time.

Associate Software Engineer August 2025 – April 2026
Todd Agriscience *Remote*

- Architected a greenfield full-stack platform with Next.js, TypeScript, Drizzle ORM, and PostgreSQL, reducing development friction through end-to-end type safety and shared validation schemas.
- Implemented GitHub Actions CI/CD pipelines and Husky hooks, eliminating 20+ build failures.
- Co-designed a 20+ table PostgreSQL schema and full-stack data architecture integrating vector search, Sanity CMS, and S3, supporting farm, crop, livestock, and soil analysis workflows.

Software Engineering Intern May 2025 – August 2025
Dwolla *Des Moines, Iowa*

- Streamlined components on a fin-tech payment platform processing $85B+ in annual payments, unifying 30+ fragmented UI instances, reducing legacy code redundancy and improving cross-feature consistency.
- **Safeguarded 1,200 users** by debugging a critical prefetching bug, preventing imminent Cloudflare IP bans.
- Designed and delivered 10+ features for a payment operations dashboard using Next.js, Redux Toolkit, and Typescript, improving fund transfer visibility and accessibility for enterprise clients.

Projects

Flutr.org – Senior Design Butterfly Management App | *Drizzle ORM, PostgreSQL, Docker, Digital Ocean*

- Increased operational efficiency by **40%** for **2+ orgs** by conducting stakeholder interviews, co-authoring a 49-page technical design document, and engineering a WCAG-compliant multi-tenant SaaS with granular RBAC.

Campus Mind – 2025 Hack Midwest | *Next.js, TypeScript, Python, FastAPI, MongoDB, GCP, AWS Strands*

- Scoped and developed a Canvas companion app with auto-syncing courses and assignments to Google Calendar, automated AI study block creation, AI health checkins, AI guide creation, and vector search assignment finder.

Czchen.dev – OS Inspired Portfolio | *Next.js, TypeScript, MongoDB, Node.js, OpenAI API, Spotify API*

- Designed an OS-inspired interactive portfolio displaying 15+ projects and 10+ custom apps, including an AI academic and wellness app, Spotify clone with audio playback, and a community-driven event finder.

Technical Skills

Languages: TypeScript, JavaScript, Java, Python, C, Bash, SQL, HTML/CSS
Frameworks: Next.js, React.js, GraphQL, Redux Toolkit, Express.js, Node.js, FastAPI, Spring Boot, Jest, JUnit
Databases/Cloud: AWS, GCP, Docker, PostgreSQL, MongoDB, Firebase, Supabase
Tools/DevOps: Git, GitHub Actions, Jira, Linux, CI/CD, RESTful APIs